

Microeconomics for Finance

Equilibrium in an Exchange Economy

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Introduction

- An exchange economy corresponds to a situation in which several agents are initially endowed with consumption goods.
- From a given basket of goods, an agent draws a certain level of “satisfaction” or **utility** which is measured by a utility function.
- Agents are allowed to trade their initial endowments with each other, in a **Walrasian market**, to possibly increase their utility.
- The efficiency of the market outcome can be assessed with the concept of **Pareto optimality**.
- In this class we focus the simplest setup : 2 agents and 2 goods.

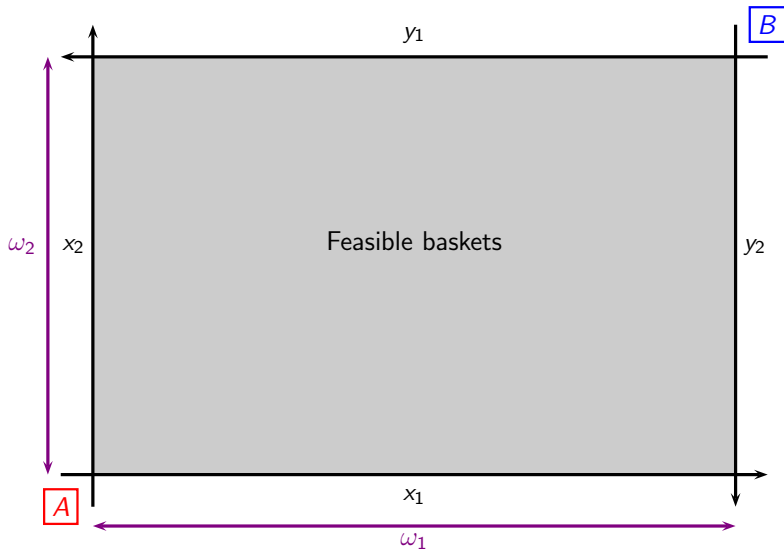
2 × 2 Economy

- Two agents, A and B , and two goods, 1 and 2.
- Initial endowments :
 - A endowed with quantities ω_1^A of good 1, and ω_2^A of good 2
 - B endowed with quantities ω_1^B of good 1, and ω_2^B of good 2
- Aggregate resources of goods 1 and 2 : $\omega_1 = \omega_1^A + \omega_1^B$, $\omega_2 = \omega_2^A + \omega_2^B$.
- Feasible baskets for A : (x_1, x_2) with $x_1 \geq 0, x_2 \geq 0$.
- Feasible baskets for B : (y_1, y_2) with $y_1 \geq 0, y_2 \geq 0$.
- Feasible allocation : a pair of baskets, $\{(x_1, x_2), (y_1, y_2)\}$, such that

$$x_1 + y_1 = \omega_1, \text{ and } x_2 + y_2 = \omega_2$$

- The **Edgeworth box** represents feasible allocations.

Edgeworth box



Walrasian Market

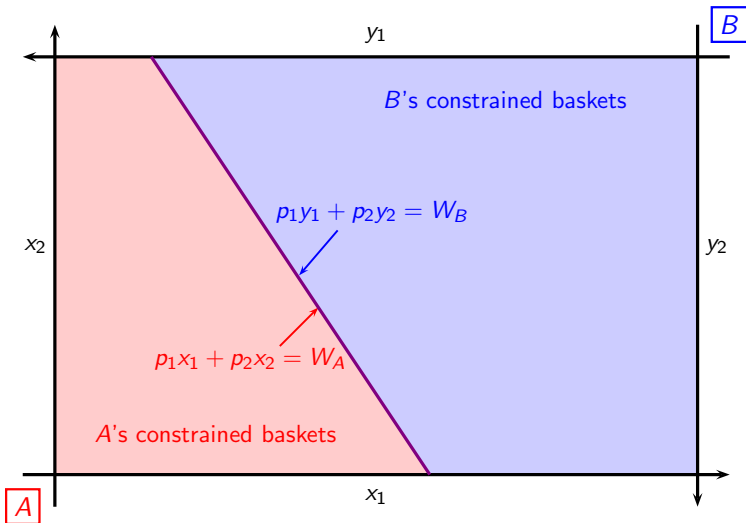
- Good 1 can be traded (bought and sold) at price p_1 . Good 2 can be traded at price p_2 .
- These prices are not parameters. Our objective is to determine the prices that clear the markets (s.t demand = supply)
- Given p_1 and p_2 , A 's initial wealth is $W_A = p_1\omega_1^A + p_2\omega_2^A$.
- Feasible baskets under budget constraints for A : (x_1, x_2) such that

$$p_1x_1 + p_2x_2 \leq W_A$$

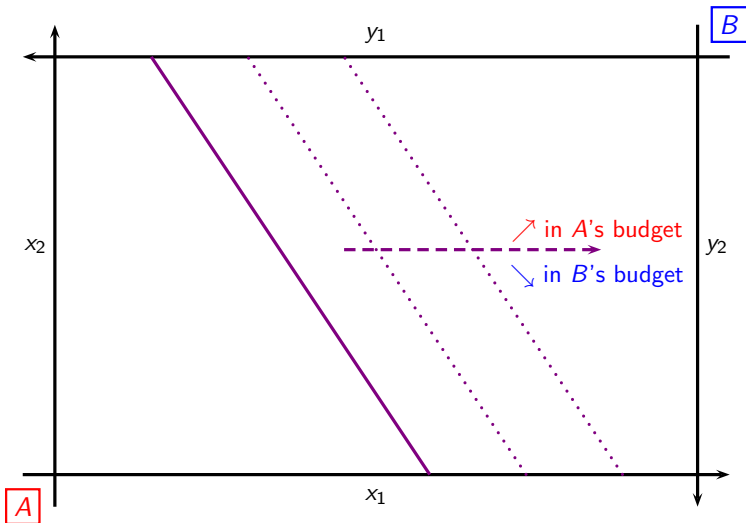
Optimally her budget is exhausted. She will pick an allocation on the *budget line* : $p_1x_1 + p_2x_2 = W_A$. Idem for B .

- Budget constraints partition the Edgeworth box.

Budget Constraints



Budget Constraints



Preferences

- Individual preferences of A over baskets represented by the utility function, u^A . A specific (feasible) basket (x_1, x_2) yields a utility level $u^A(x_1, x_2) \in \mathbb{R}$. Idem for B .
- The utility function should be strictly concave, and strictly increasing in each good, that is

$$\frac{\partial u^A}{\partial x_1} > 0, \quad \frac{\partial u^A}{\partial x_2} > 0, \quad \frac{\partial^2 u^A}{\partial x_1^2} < 0, \quad \frac{\partial^2 u^A}{\partial x_2^2} < 0.$$

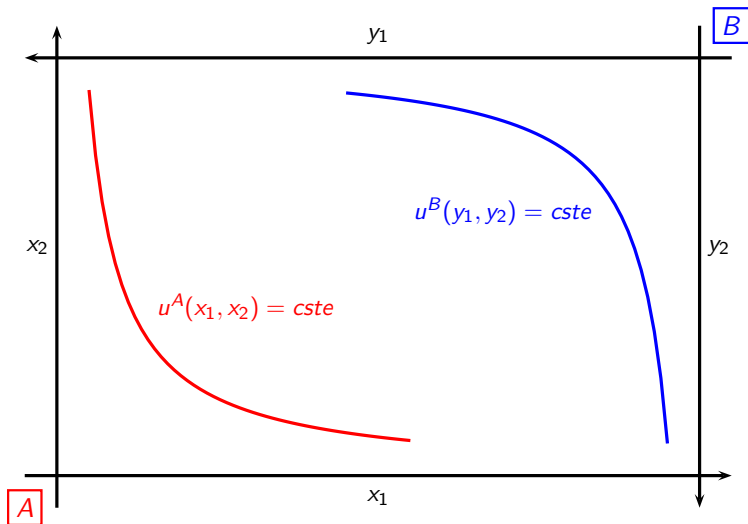
- Example : Cobb-Douglas utility function

$$u^A(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}, \quad u^B(y_1, y_2) = y_1^\beta y_2^{1-\beta},$$

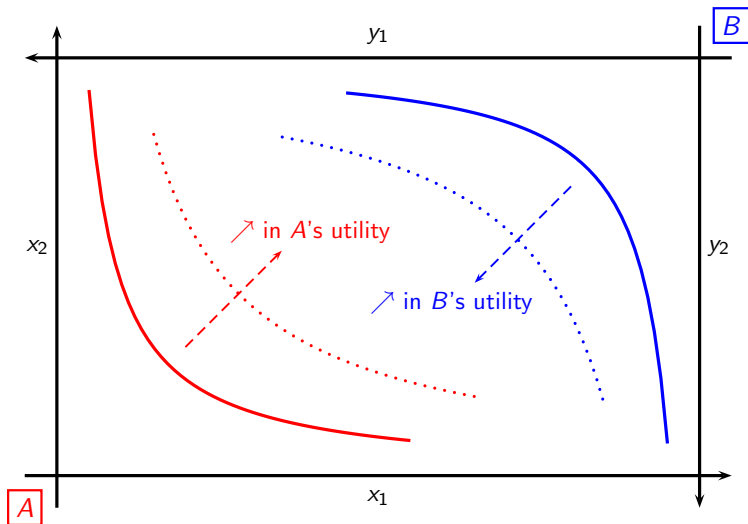
with $\alpha \in [0, 1]$, $\beta \in [0, 1]$.

- One can represent iso-utility curves, that sets of baskets generating a given utility level, in the Edgeworth box

Iso-utility Curves



Iso-utility Curves



Pareto Optimality

- Allocation $\{(x_1, x_2), (y_1, y_2)\}$ dominates allocation $\{(x'_1, x'_2), (y'_1, y'_2)\}$ in the Pareto sense if

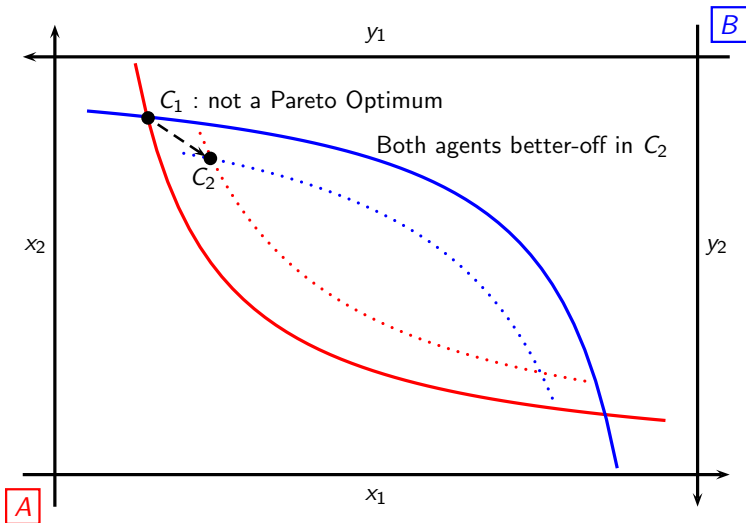
$$u^A(x_1, x_2) \geq u^A(x'_1, x'_2), \text{ and } u^B(y_1, y_2) \geq u^B(y'_1, y'_2),$$

with at least one strict inequality,

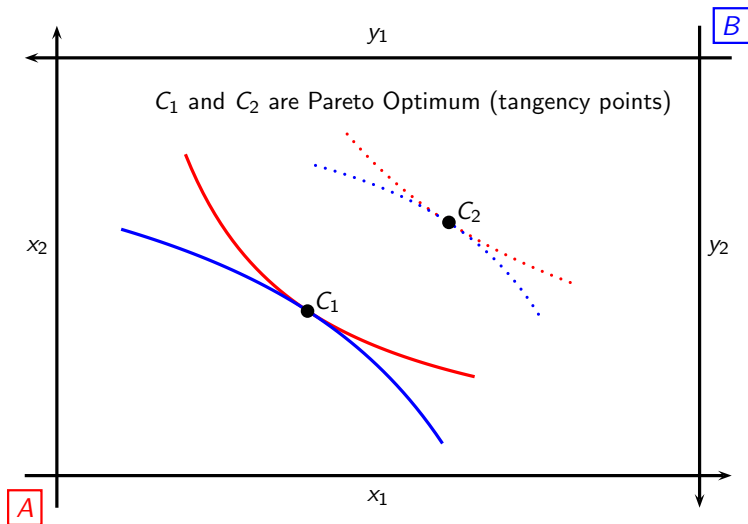
that is if both agent are better off with the first allocation.

- A feasible allocation $\{(x_1, x_2), (y_1, y_2)\}$ is *Pareto optimal* if it is not dominated by any feasible allocation in the Pareto sense.
Put differently, any other feasible allocation would make (at least) one agent worse off.
- Typically, there are multiple (a set of) Pareto optimal allocations.

Pareto Optimality



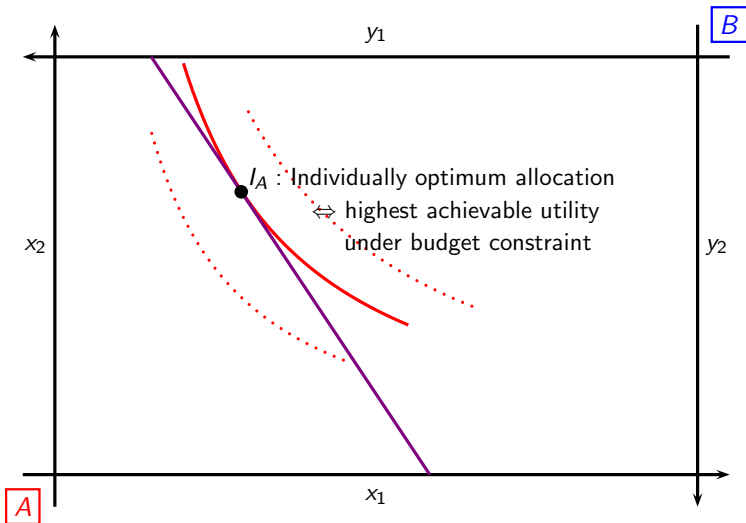
Pareto Optimality



Individual Optimization

- **Result :** Under the strict concavity and increasing assumptions on u^A , and for given prices p_1 and p_2 , there exists a unique *optimal* basket (x_1, x_2) . It maximizes $u^A(x_1, x_2)$ under the budget constraint.
- Based on this existence and uniqueness result, one can define the demand function of A .
- The demand function of A maps each set of price (p_1, p_2) to the optimal basket, $(x_1(p_1, p_2), x_2(p_1, p_2))$.
- Idem for B .

Individual Optimization



Individual Optimization

- The constrained optimization program of A can be written as :

$$\max_{x_1, x_2} u^A(x_1, x_2) \text{ under constraint (u.c) } p_1 x_1 + p_2 x_2 = p_1 \omega_1^A + p_2 \omega_2^A$$

- In the 2×2 economy, this multi-dimensional optimization problem can be turned into a one-dimensional one. Use the budget constraint to define x_2 as a function of x_1 :

$$x_2 = \frac{p_1}{p_2} \omega_1^A + \omega_2^A - \frac{p_1}{p_2} x_1.$$

- The optimization program becomes

$$\max_{x_1} u^A \left(x_1, \frac{p_1}{p_2} \omega_1^A + \omega_2^A - \frac{p_1}{p_2} x_1 \right)$$

- It is solved by taking the *first order condition (F.O.C)*,

$$\frac{\partial}{\partial x_1} \left\{ u^A \left(x_1, \frac{p_1}{p_2} \omega_1^A + \omega_2^A - \frac{p_1}{p_2} x_1 \right) \right\} = 0,$$

to obtain directly the optimal $x_1(p_1, p_2)$, and then $x_2(p_1, p_2)$.

Individual Optimization

- The standard optimization technique is the **Lagrangian method**.
- One defines the Lagrangian of the constrained optimization program :

$$\mathcal{L}^A = u^A(x_1, x_2) + \lambda^A(p_1\omega_1^A + p_2\omega_2^A - p_1x_1 - p_2x_2)$$

where λ^A is a new variable called the *Lagrange multiplier*.

- The Lagrangian (and its multiplier) is specific to A . B has his own.
- It is solved by taking three F.O.C's,

$$\frac{\partial \mathcal{L}^A}{\partial x_1} = 0, \quad \frac{\partial \mathcal{L}^A}{\partial x_2} = 0, \quad \text{and} \quad \frac{\partial \mathcal{L}^A}{\partial \lambda^A} = 0$$

(the last one returns the budget constraint).

Individual Optimization

- Taking the two first F.O.C's of the Lagrangian yields

$$\frac{\partial u^A}{\partial x_1} = \lambda^A p_1, \text{ and } \frac{\partial u^A}{\partial x_2} = \lambda^A p_2$$

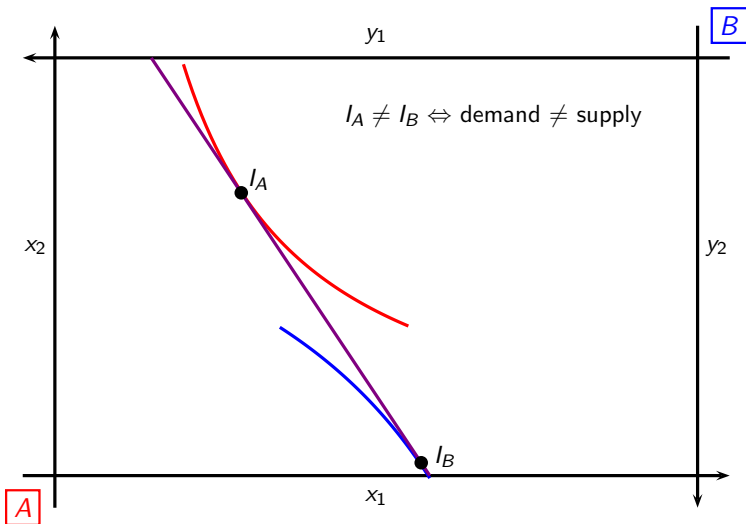
- Taking the ratio yields :

$$\frac{\frac{\partial u^A}{\partial x_1}}{\frac{\partial u^A}{\partial x_2}} = \frac{p_1}{p_2}$$

That is a “*slope of the iso-utility curve = slope of the budget line*” condition.

- As the optimal allocation should be on the budget line, it is uniquely pinned-down as the tangency point between an iso-utility curve and the budget line.

Individual Optimization



Walrasian Equilibrium

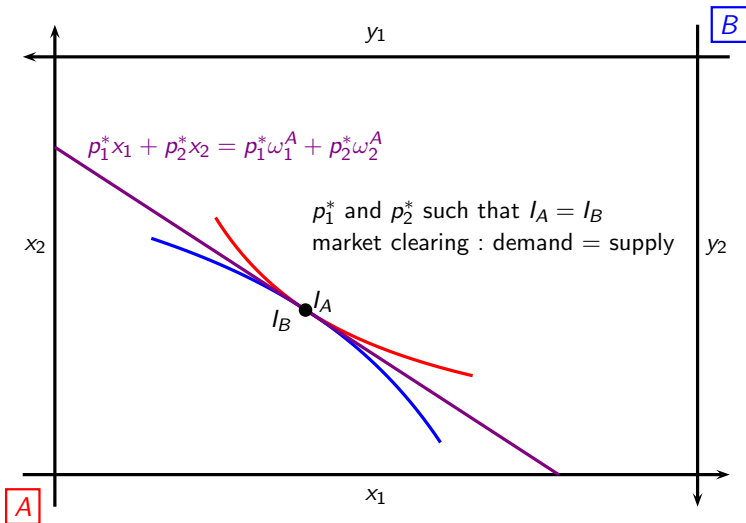
Definition. An (Walrasian) equilibrium is $\{(x_1^*, x_2^*), (y_1^*, y_2^*), p_1^*, p_2^*\}$ such that

- ① (x_1^*, x_2^*) maximizes $u^A(x_1, x_2)$ u.c $p_1^* x_1 + p_2^* x_2 = p_1^* \omega_1^A + p_2^* \omega_2^A$
- ② (y_1^*, y_2^*) maximizes $u^B(y_1, y_2)$ u.c $p_1^* y_1 + p_2^* y_2 = p_1^* \omega_1^B + p_2^* \omega_2^B$
- ③ market clearing : $x_1^* + y_1^* = \omega_1$ and $x_2^* + y_2^* = \omega_2$

Equilibrium prices are not uniquely determined - only up to a scalar $\Rightarrow$ one can choose a numeraire, for instance $p_2^* = 1$.

First Welfare Theorem. If $\{(x_1^*, x_2^*), (y_1^*, y_2^*), p_1^*, p_2^*\}$ is an equilibrium then $\{(x_1^*, x_2^*), (y_1^*, y_2^*)\}$ is a Pareto optimal allocation.

Walrasian Equilibrium



Further (Important) Results

- **Second Welfare Theorem.** If $\{(x_1^*, x_2^*), (y_1^*, y_2^*)\}$ is a Pareto optimal allocation, then it is the allocation of an equilibrium with transfers.
- Equilibrium + transfer = take 100€ from A , give it to B , and let them trade out of their endowments + transfers.
- It means that taxes allow to reach the Pareto optimum we wish among the many.
- All the results can be generalized to an economy with $N \geq 2$ agents and $L \geq 2$ goods.